

Arman Hansda

+91 9798595716 | armanhansda46@gmail.com | GitHub | LinkedIn

SUMMARY

Mathematics & Computing student at BIT Mesra with experience building scalable full-stack and distributed systems. Proficient in Rust, Node.js, and Next.js, with hands-on experience in designing search engines, real-time systems, and high-performance backend architectures.

EDUCATION

Birla Institute of Technology, Mesra

Integrated M.Sc. in Mathematics and Computing

GPA: 8.4 / 10

Sept 2023 – present

TECHNICAL SKILLS

Languages: C++, Rust, JavaScript, TypeScript, Python

Frameworks & Libraries: React.js, Next.js, Express.js, tRPC, TailwindCSS

Backend & Systems: Node.js, REST APIs

Databases & Caching: PostgreSQL, MongoDB, Redis

Tools & DevOps: Git, GitHub, Vercel, Cloudinary

Core Concepts: Data Structures & Algorithms, OOP, System Design, Operating Systems, DBMS, Distributed Systems, Authentication

PROJECTS

ArachNode – Search Engine (Rust, NextJs, NodeJs, RESTapi, Redis Queue)

GitHub

- Engineered a **distributed web crawler** in Rust with concurrent workers, crawling up to **10K+ pages** across domains.
- Designed **fault-tolerant URL scheduling** using Redis (queue, visited set, seen set), reducing duplicate crawling by **60%**.
- Built an end-to-end **search engine pipeline** (crawling → parsing → indexing → querying) handling **thousands of documents**.
- Implemented **inverted indexing with tokenization**, enabling sub-second keyword search over **10K+ indexed pages**.
- Developed **RESTful search APIs (Node.js)** with average response latency under **200ms**.
- Structured the system with modular components, improving scalability and maintainability for future distributed expansion.

PIXO — Full Stack Content Platform

GitHub | [Live Demo](#)

- Built a full-stack blogging platform using **Next.js, tRPC, PostgreSQL, and Drizzle ORM**, supporting **100+ users** and **500+ posts**.
- Designed type-safe APIs with **tRPC**, reducing runtime API errors by **40%**.
- Optimized PostgreSQL queries using indexing, improving response time by **30%**.
- Implemented **JWT authentication and Google OAuth**, securing **100% of protected routes**.
- Integrated **Cloudinary** for media uploads, handling **1000+ images** efficiently.
- Developed a responsive UI using **Next.js App Router and TailwindCSS**, improving page load performance by **25%**.

SnowID – Distributed Unique ID Generator

GitHub

- Built a **60-bit distributed ID generator** supporting **64 nodes** and generating **8K IDs/ms per node**.
- Achieved **1M IDs/sec throughput** with compact **Base62 (10-character) encoding**.
- Reduced ID size by **40% compared to UUID**, improving storage and transmission efficiency.
- Implemented **clock drift handling with monotonic guarantees**, ensuring consistency in distributed systems.
- Developed **benchmarking and decoding tools**, validating performance under high-load scenarios.